absence of bias, while a value close to 1 suggests a high degree of bias. Conversely, a value of -1 would indicate the presence of a reverse bias, where the evaluator f tends to undervalue its own responses. In this definition, the case where y_1 is preferred is explicitly considered, but this does not result in a loss of generality by treating the preferred response as y_1.

4.2 Experimental Setting

The aim of our experiment is to quantify the self-preference bias in various LLMs. To address a wide range of tasks and topics, we have LLMs evaluate responses in open-ended dialogues and measure the bias using the Definition 4.1.

First, we provide the LLM evaluator with a user query written by a human and two responses generated by different LLMs. We then ask the LLM evaluator to determine which of the two responses demonstrates higher quality. For clarity and ease of reference during evaluation, we designate the two responses as "response A" and "response B". Finally, using the probabilities that the LLM evaluator outputs for "A" and "B", denoted as $p(\text{A}|\text{context})$ and $p(\text{B}|\text{context})$, we calculate the score for response A using the following equation.

$$\text{score}_{\text{A}} = \frac{p(\text{A}|\text{context})}{\sum_{w \in \{\text{A, B}\}} p(w|\text{context})} \tag{3}$$

The score for response B is calculated in the same manner. This post-processing, as shown in Equation 3 is designed to enhance the interpretability of analyses across LLMs with different probability distributions, following the approach of Schick et al. [2021]. Additionally, according to Zheng et al. [2024], LLMs may exhibit position bias, which refers to the tendency to prefer responses located in specific positions within the prompt. To mitigate this bias, we swap the positions of responses A and B, and the evaluation scores are averaged over two iterations. Using the scores obtained through the above process, we examine the extent to which there is a difference in evaluation scores between the LLM's own generated response and responses generated by other LLMs.

Empirical evaluation uses Chatbot Arena dataset [Zheng et al., 2024], which contains 33,000 dialogues and each consisting of a user query and a pair of responses generated by two different LLMs. We calculate evaluation scores for the pre-existing response pairs stored in the dataset. In this dataset, every response pair is labeled with either "model_a", "model_b", or "tie" as a result of a human evaluation comparing the quality of the responses.

As the LLM evaluators, we employed the following eight LLMs, which also used in Chatbot Arena dataset: the closed-source GPT-3.5-Turbo and GPT-4 [Josh et al., 2024], and the open-source Vicuna-7b, Vicuna-13b [Chiang et al., 2023], oasst-pythia-12b [Biderman et al., 2023], dolly-v2-12b [Conover et al., 2023], Koala-13b [Geng et al., 2023], and stablelm-tuned-alpha-7b [Jonathan Tow, 2023]. We used the same prompt as Zheng et al. [2024] to calculate the evaluation scores of responses A and B for all LLM evaluators. This prompt was created based on the work of Zheng et al. [2024], and it instructs the LLM to output either "[[A]]", "[[B]]", or "[[C]]" after providing an explanation. In the implementation, we used the output probability distribution following the token corresponding to "[[" in the generated text to compute the score using Equation 3. Additionally, if the LLM evaluators failed to comply with the prompt instructions and did not provide a response in the format of "[[A]]" or a similar structure, these cases were excluded from the analysis.

4.3 Result

The results of the bias measurement using Definition 4.1 are presented in Figure 1b. It was confirmed that GPT-4 exhibits the highest self-preference bias. Definition 4.1 focuses on the recall of the LLM evaluator concerning both high and low ratings by the human evaluator. Thus, it can be concluded that GPT-4 showed lower recall in cases where humans evaluated unfavorably compared to when higher evaluating. When examining the recall values in the confusion matrix shown in Figure 2, they are calculated as $0.945 \approx \frac{1852}{108+1852}$ and $0.425 \approx \frac{118}{160+118}$. The difference between these values is 0.520, which corresponds to the value reported in Figure 1b. Following GPT-4, Vicuna-13b and Koala-13b also exhibited significant bias. In contrast, other LLMs displayed values relatively close to zero. Notably, oasst-pythia-12, dolly-v2-12b, and stablelm-tuned-alpha-7b showed negative values, indicating a reverse bias where the LLMs tend to underestimate their own outputs.

Table 1 presents a randomly selected example of self-preference bias in GPT-4, where humans favored the alternative response. In this example, the user query is a straightforward request to list blue items. While GPT-4 states that it lacks physical recognition before listing, GPT-3.5-Turbo directly lists the blue items without any such explanation. Both responses are of high quality, and the final evaluation reflects the evaluator's policy and stylistic preferences. Although humans and GPT-3.5-Turbo preferred the response from GPT-3.5-Turbo, GPT-4 favored its own response, illustrating a typical case of self-preference bias.

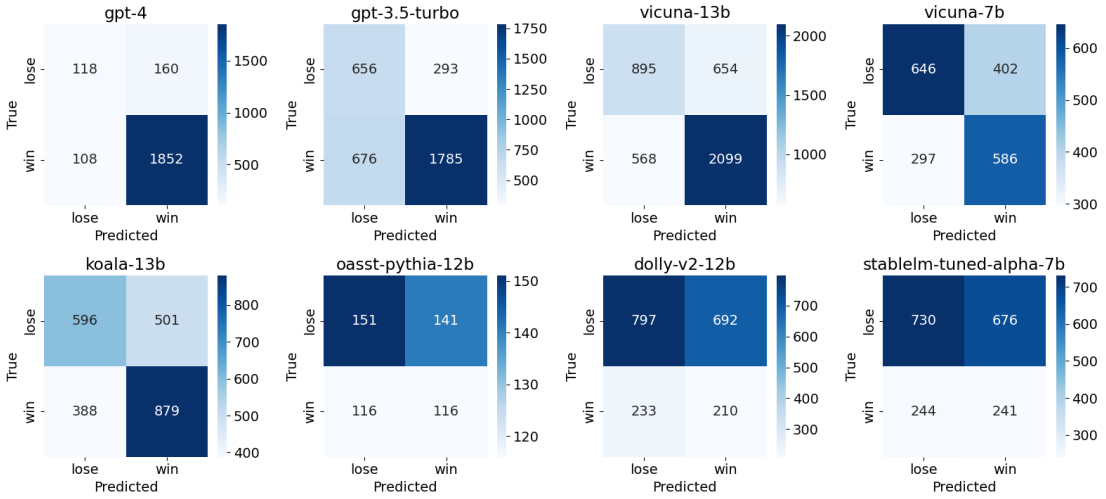

Figure 2: Confusion matrix for each LLM evaluator's assessment of pairs including its own output. It suggests that **some LLMs, including GPT-4, have relatively high true positive rates than true negative rate, which means that they have self-preference bias.**

Table 1: **An example of self-preference bias in GPT-4.** The user query involves a simple request to list blue items. GPT-4 responds by first acknowledging its lack of physical cognition, and then proceeds to list blue items. In contrast, GPT-3.5-Turbo directly lists blue items without any preliminary explanation which human evaluation prefers.

User: Please list the blue items	
GPT-4: As a language model AI, I can't see the physical objects around you. However, here are some common blue items that people might have: 1. Blue shirt 2. Blue jeans 3. Blue pen 4. Blue coffee mug 5. Blue pillow 6. Blue blanket 7. Blue towel 8. Blue notebook 9. Blue umbrella 10. Blue vase Please specify the context or the category of items you are interested in if you need a more precise list.	**GPT-3.5-Turbo**: Sure, here are some blue items: 1. Blue jeans 2. Blueberries 3. Blue shirt 4. Blue pen 5. Blue car 6. Blue backpack 7. Blue notebook 8. Blue umbrella 9. Blue sneakers 10. Blue curtains
GPT-4 preferred	**Human and GPT-3.5-Turbo preferred**

5 How Do LLMs Overestimate Their Own Outputs?

We have shown that certain LLM evaluators tend to assign disproportionately high scores based on whether the output being generated from themselves, even though LLM evaluators are not explicitly provided with labels indicating whether a given output is their own. We hypothesized that LLM evaluators might be affected by the similarity of the response to their own output. To investigate this further, we focus on how LLM evaluators change their evaluation depending on the perplexities of responses.

As in previous experiments, we employed the pairwise evaluation framework where evaluators compare two responses, response A and B. First, we computed the perplexities of responses conditioned on the prompt, and took the difference between response A and B for all samples. Next, we divided the perplexity differences into bins and calculated the probabilities that each LLM evaluator judged response A as the winner in each subset. Additionally, we computed the winning judgment rate of the human for response A within each bin. In this experiment, we excluded GPT-4 and GPT-3.5-Turbo, as perplexity values could not be obtained for these models.

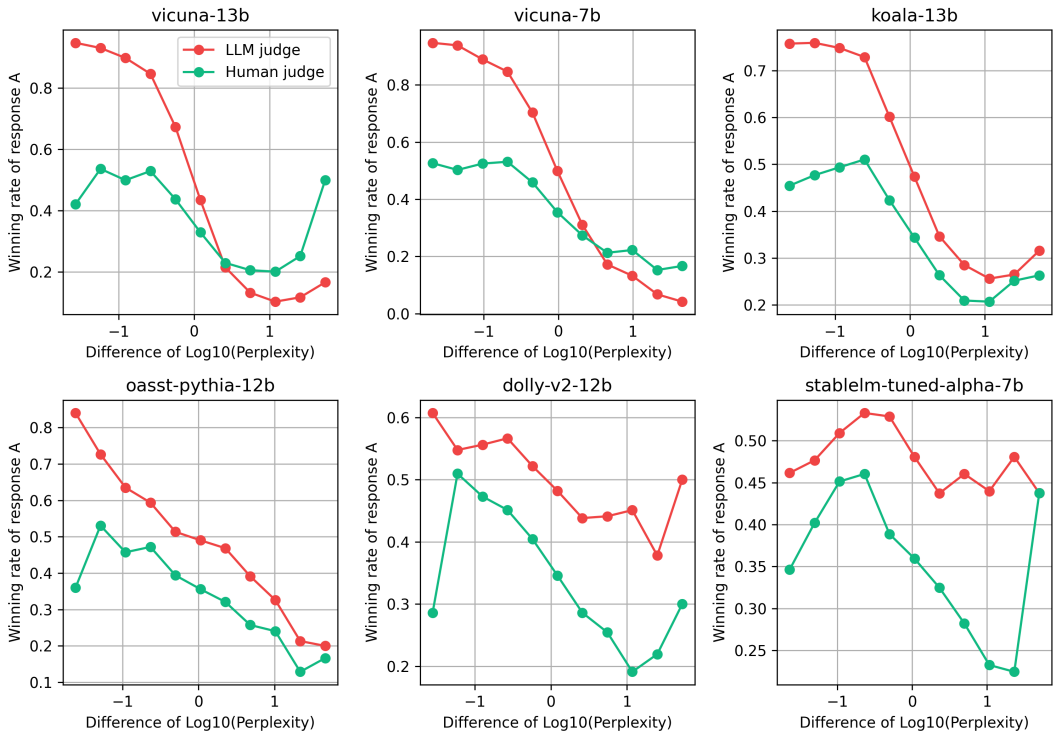

Figure 3: **LLMs vs human conditioned on perplexity.** Winning judgment rates by LLMs conditioned on perplexity with human winning judgment rates are plotted. All models except dolly-v2-12b and stablelm-tuned-alpha-7b demonstrated a clear tendency to assign higher evaluations to responses with lower perplexity.

Figure 3 shows the comparison of winning judgment rates within each perplexity bin for six models. All models except stablelm-tuned-alpha-7b demonstrated a clear tendency to assign higher evaluations to responses with lower perplexity. Furthermore, it was confirmed that this tendency was stronger in vicuna-13b, vicuna-7b, koala-13b, and oasst-pythia-12b than in humans. This result indicates that LLM evaluators overly change their evaluation depending on perplexities of responses.

To further investigate the effect of differences in whether the output is self-generated or not, we segmented the evaluation results by the LLM evaluators into two distinct groups: one where the LLM evaluators' own output is included in the pair and another where it is not. We present the results with the output of the LLM evaluator as response A, without loss of generality. As shown in Figure 4, the winning judgment rates between two groups were similar for all models except dolly-v2-12b and stable-tuned-alpha-7b. This suggests that the factor influencing the LLM evaluators' judgments is not whether the response is their own but rather the perplexity of the responses. As confirmed by Figure 5, LLMs tend to exhibit lower perplexity for their own outputs. In other words, these findings imply that self-preference bias may be a phenomenon where the model's own output inherently exhibits lower perplexity.

In these experiments, we were unable to obtain perplexity values from GPT-4 and GPT-3.5-Turbo, resulting in a lack of analysis on the competitive LLMs. To address this gap, we conducted additional experiments using Llama2 [Touvron et al., 2023] and Llama3 [Dubey et al., 2024]. However, the Chatbot Arena dataset does not contain human annotations for these LLMs' responses. Therefore, we obtained evaluations for existing responses from these Llama models and only compared them with human evaluations.

The results are presented in Figure 6. We found that all models changed their evaluations more than humans, depending on the difference in perplexity. This indicates that even in the competitive models, including Llama 3.1, perplexity may be causing unfair bias in the evaluation.

6 Discussion

To reduce self-preference bias, one possible approach is ensemble evaluation using multiple models. This method is expected to provide a more equitable evaluation by avoiding reliance on a single model. Specifically, when a model